English-speaking nine-year-olds). It is named after the ancient sky god who, according to Greek myth, was Saturn's father and thus the grandfather of the Olympian gods.

We no longer consider the Sun and Moon to be planets, and—ignoring the comparatively insignificant asteroids and comets—count Uranus as the seventh planet in order from the Sun (Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune, Pluto). It is the first planet unknown to the ancients. The four outer, Jovian, planets turn out to be very different from the four inner, terrestrial, planets. Pluto is a separate case.

As the years passed and the quality of astronomical instruments improved, we began to learn more about distant Uranus. What reflects the dim sunlight back to us is no solid surface, but atmosphere and clouds—just as for Titan, Venus, Jupiter, Saturn, and Neptune. The air on Uranus is made of hydrogen and helium, the two simplest gases. Methane and other hydrocarbons are also present. Just below the clouds visible to Earthbound observers is a massive atmosphere with enormous quantities of ammonia, hydrogen sulfide, and, especially, water.

At depth on Jupiter and Saturn, the pressures are so great that atoms sweat electrons, and the air becomes a metal. That does not seem to happen on less massive Uranus, because the pressures at depth are less. Still deeper, discovered only by its subtle tugs on Uranus' moons, wholly inaccessible to view, under the crushing weight of the overlying atmosphere, is a rocky surface. A big Earthlike planet is hiding down there, swathed in an immense blanket of air.

The Earth's surface temperature is due to the sunlight it intercepts. Turn off the Sun and the planet soon chills—not to trifling Antarctic cold, not just so cold that the oceans freeze, but to a cold so intense that the very air precipitates out, forming a ten-meter-thick layer of oxygen and nitrogen snows covering the whole planet. The little bit of energy that trickles up from the Earth's hot interior would be insufficient to melt these snows. For Jupiter, Saturn, and Neptune it's different. There's about as much heat pouring out from their interiors as they acquire from the warmth of